

Harnessing Artificial Intelligence for Innovation Environment: Implications for Startup Enterprises' Performance in the Kingdom of Saudi Arabia

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Abstract

In alignment with Saudi Vision 2030, this study examines the role of artificial intelligence (AI) as a catalyst for research and development (R&D) enhancement in Saudi startups, focusing on the mediating effect of the innovation environment, which encompasses resource availability, leadership support, risk culture, learning orientation, and external collaboration. Using a PRISMA 2020-guided systematic synthesis of 220 peer-reviewed studies and policy documents (2020–2025), we demonstrate that AI adoption drives 25% higher productivity and improves decision-making, problem-solving, and customer satisfaction. However, talent shortages, fragmented institutional support, and underdeveloped infrastructure critically hinder scalability. This study advocates empirically grounded policies, academic-industry partnerships, and targeted investments to enable the sustainable deployment of AI. By empirically establishing the mediating mechanism of the innovation environment, this work fills a critical gap in the literature and advances Saudi Arabia's ambitions for a knowledge-based economy.

Keywords:

Artificial Intelligence, Research and Development, Startup Enterprises Performance, Innovation Environment, and the Kingdom of Saudi Arabia

Introduction

This paper examines the impact of artificial intelligence (AI) on the research and development (R&D) activities of startup companies in Saudi Arabia, with a particular focus on the Vision 2030 initiative. AI is being acknowledged as a strategic resource that promotes productivity, innovation, and sustainable economic growth, and this integration of AI and deep tech into the R&D functions of startups is entirely in line with the Kingdom's national agenda towards a better diversified economy increasingly based on a knowledge society, rather than on oil and gas exploitation. While its potential is enormous, small and medium-sized enterprises in Saudi Arabia face several major impediments, including limited access to human capital with the proper skill set, inadequate digital infrastructure, and uncertainty regarding consumer preparedness. Such limitations hinder their ability to successfully adopt, embrace, and deploy AI through innovative and competitive practices in both local and global markets.

AI, as a disruptive phenomenon, is swiftly embedding itself within startup ecosystems, altering the landscape of global entrepreneurship and redefining norms in innovation and contention models. This transformation is further catalysed in the Kingdom of Saudi Arabia (KSA) through Vision 2030, which places a strategic focus on economic diversification through AI, and perceives startups as a vital vehicle to facilitate technology delivery (SDAIA, 2020); Attention to the gradualness of AI adoption underscores the extent to which AI adoption contributes to improved startup performance is conditional on the multidimensional nature of Innovation Environment (IE), which includes policy frameworks, resource availability, and collaboration (Hossain, 2024). Although research on AI startup growth-triggered progression (Bati et al., 2021) and the generic impact of IE has surfaced, very few studies have investigated the culturally and institutionally specific dimensions of IE and whether these dimensions mediate how AI works in Saudi Arabia, where the environment for entrepreneurship is significantly different.

This research presents a practical framework for AI adoption in the startup ecosystem, thereby filling a gap in the current literature. It highlights the need for focused strategies to address the challenges faced by these companies, especially when it concerns the use case of AI in R&D. Through AI, the ability to make optimal choices, boost productivity, and carry out faster product development cycles is now becoming attainable at all operational levels. This research is designed to assess the implications of AI not only on individual businesses but also on the innovation ecosystem in Saudi Arabia. It thus delivers evidence-based insights that can guide policy, inform entrepreneurs, and contribute to the long-term national goals of economic transformation and technological leadership.

Background of AI integration in research and development (R&D)

As part of its Vision 2030, an ambitious national framework to diversify its economy and develop its technological base, Saudi Arabia is focusing on Artificial Intelligence (AI) as a source of growth especially in R&D. These investments in AI are predicted to generate innovation, economic growth, job creation, and bring in even more needs within a short period. The rapid growth of data creation and the advancement of computing power have enabled the utilization of AI, particularly in areas such as speech recognition, text analysis, vision, healthcare, communication, finance, and manufacturing. In this changing context, startups have played an important role in local economic development, particularly in terms of research and development, that is, R&D efforts; AI is integrated into the economy, it is projected to lead to a 10–15% rise in national productivity, equivalent to approximately SAR 500 billion, according to government projections (Mashat, 2020; Saudi Vision 2030, 2021).

Research Purpose and Analytical Approach

The objectives are to study how the integration of AI in R&D processes affects the performance of start-up enterprises within Saudi Arabia's innovation environment, in the context of Vision 2030. One of the main boxes in the framework is the mediating role of the Innovation Environment, which follows the expectation that the Innovation Environment mediates the relationship between AI capabilities and business performance; The research aims to give based solutions for entrepreneurs, policymakers, and investors on how to deal with hurdles in the way of adoption and create a high-performing startup ecosystem and Startups boost innovation, create jobs, and support economic diversification in line with Saudi Arabia's Vision 2030 agenda; Nevertheless, these enterprises face considerable challenges in implementing Artificial Intelligence (AI) within their Research and Development (R&D) functions. AI holds the potential to create powerful avenues for better decision-making, faster innovation cycles, and healthier

productivity levels. Still, most Saudi start-ups fail to harness the technology due to a lack of financial capital, human resources, and digital infrastructure. As evidence, only 15% of Saudi startups receive sufficient AI funding (Alshamrani, 2022); 60% face a scarcity of an AI-skilled workforce (SDAIA, 2020); and the synergy between academia and industry is limited (Bresciani et al., 2021). Consequently, the national AI market, with an expected compound annual growth rate (CAGR) of 43.1%, is not fully leveraged by Saudi startups. At the same time, the level of AI adoption remains far lower than in the economies of developed countries.

Institutional Support and Innovation Challenges

To institutionalize and accelerate innovation, the Kingdom has established a research and development (R&D) governance structure in line with Vision 2030. The establishment of the Supreme Committee for Research, Development, and Innovation in 2021 and the subsequent launch of the Research, Development, and Innovation Authority (RDIA) later that year represent foundational steps towards regulating and promoting strategic R&D programs; These associations support leading initiatives and distribute resources aligned with Saudi Arabia's transformation goals. Despite this institutional progress, startups continue to face challenges such as limited funding, a shortage of AI-skilled professionals, and inadequate collaboration between academia and industry.; and these obstacles hinder their ability to fully leverage AI technologies for R&D and innovation, which are crucial for achieving economic competitiveness and a knowledge-based economy (RDIA, 2023; Alshamrani, 2022; Bresciani et al., 2021).

Objectives and Questions Research

To achieve its aim, the study is guided by four key research objectives: (RO-1) to assess the impact of AI-driven organizational capabilities and the innovation environment. (RO-2) to explore the influence of the innovation environment on startup performance; and (RO-3) to investigate whether the innovation environment mediates the relationship between AI capabilities and startup outcomes. These objectives are translated into three corresponding research questions: (RQ-1) What is the relationship between AI-driven organizational capabilities and the Innovation Environment? (RQ-2) What is the impact of the innovation environment on Startup Enterprise Performance? (RQ-3) How does the Innovation Environment mediate the relationship between AI-driven Organizational Capabilities (AIOC) and Startup Enterprise Performance in Saudi Arabia?; The answers to these questions will not only address existing empirical gaps but also offer practical frameworks for advancing AI-based innovation within Saudi Arabia's startup landscape.

Significance Contribution

This study holds both practical and theoretical significance in understanding the role of AI in transforming the startup sector in Saudi Arabia. Practically, it provides a roadmap for entrepreneurs to optimize AI adoption strategies that enhance decision-making, operational efficiency, and customer satisfaction. Additionally, the study offers policymakers insights into measuring and evaluating AI-driven incentives and capacity-building programs by applying them to an underexplored emerging market; It addresses literature gaps by examining how AI capabilities in startups improve productivity, accelerate innovation cycles, and foster market adaptability, primarily when supported by a conducive innovation environment (Dwivedi et al., 2020; Zhang et al., 2024).

Scope and Research Design

The study examines the intersection of artificial intelligence (AI), innovation environments, and startup enterprise performance within the specific context of Saudi Arabia's evolving entrepreneurial ecosystem; The review focuses on peer-reviewed studies and authoritative reports published between 2020 and 2025, coinciding with the implementation period of Saudi Vision 2030, which has significantly shaped the nation's innovation landscape.

The analysis concentrates on three core dimensions of the innovation environment: policy frameworks (including SDAIA regulations and RDIA funding mechanisms), resource availability (particularly AI talent pools and technological infrastructure), and collaboration networks (such as university-industry partnerships). These dimensions are explicitly examined in terms of their impact on AI adoption and startup performance within technology-driven sectors, including fintech, healthcare technology, and education technology; The review excludes non-technical applications of AI and focuses

exclusively on registered startups in their early to growth stages 1-5 years, while omitting larger corporations and traditional SMEs without an R&D focus.

The study investigates the impact of integrating AI into R&D on the performance of startup enterprises within Saudi Arabia's innovation environment. Geographically, the primary focus is on Riyadh, Saudi Arabia, and the review encompasses this area; also, it evaluates AI-driven organizational capabilities, decision-making, problem-solving, productivity, customer satisfaction, and technological infrastructure, as well as their impact on the success of startups. The research targets startup enterprises in the finance, healthcare, Information technology, and education sectors, identified from over 656,000 registered enterprises in Riyadh city, Saudi Arabia. An approach is employed to evaluate the mediating role of the innovation environment (General Authority for Statistics, 2020). This cross-sectional analysis provides timely and localized insights that support Saudi Arabia's broader strategy for AI-enabled economic transformation.

Literature Review

The Concept of Artificial Intelligence (AI) and Interpretation

AI, or artificial intelligence, is a branch of computer science that utilizes a combination of computation and the imitation of human thought processes to address complex real-world issues. AI, as defined by Copeland (2024), is the ability of a machine to perform functions commonly associated with human intelligence, such as reasoning, decision-making, and learning. It is an interdisciplinary field that encompasses computer science, mathematics, psychology, neuroscience, and philosophy. According to Naseeb (2020), AI refers to systems in the fields of natural language processing, robotics, and computer vision that utilize environmental data to operate independently.

Deep learning, neural networks, and reinforcement learning have rapidly evolved as the core technologies for artificial intelligence (AI), enabling AI to scale and make real-time decisions across various domains (Zhang & Lu, 2021) and combined with other technologies like cloud computing, IoT and blockchain, AI facilitates the creation of scalable and innovative systems capable of helping economic and societal transformations. Various such advancements have been invested in by companies such as Google and IBM, further establishing AI as a foundation for innovations.

Evolution and advancements in AI technologies

AI has made impressive strides in the past few years, and the idea of imitating human thought in the mid-20th century led to the development of rule-based systems in the 1970s and machine-learning algorithms in the 1980s. Both appeared hopeful, but extensive AI development occurred in the 1990s, when neural networks regained popularity under the framework of deep learning (DL). This development has prompted a wave of innovations that have enabled cutting-edge natural language interpretation, computer vision, and self-driving cars.

The factors behind advancement in AI include large datasets, refined algorithms, and an exponential increase in computing power. Such an achievement is viable through modern AI systems that leverage deep learning technology and reinforcement learning. As a result, the accuracy increases with numerous iterations of learning from the data. This enables AI to make a remarkable difference in various sectors, including transportation and medicine, by conducting operations that approach human-level performance.

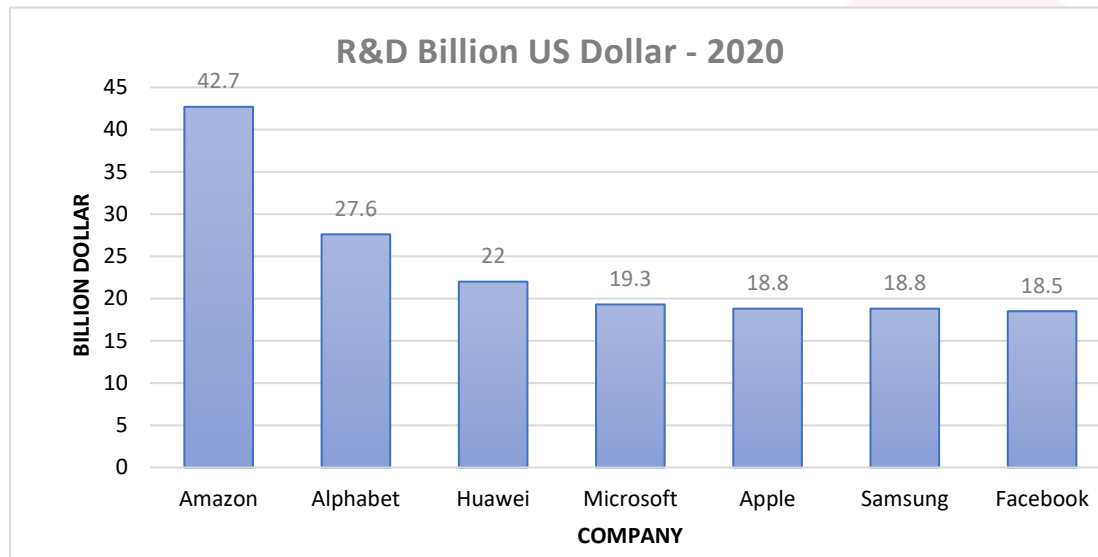
The further development of AI technologies contributes to broadening applications and the emergence of novel opportunities and avenues for innovation in numerous fields; Therefore, as the planet's economy develops in this direction, there is growing concern about the role of comprehensive regulatory and ethical frameworks enabling proper control and managing AI's profound impact on society.

Importance and role of R&D in driving innovation

Research and Development (R&D) serve as the foundation of innovative environments. Research and development expenditures are closely correlated with economic development and technological progress; furthermore, Ivanic et al. (2021) and Kenton (2024) explain that R&D expenditures serve as both a measure and a driver of innovation output, including patents and product development. Robust R&D investment enables firms to commercialize new ideas, improve product quality, and enhance national competitiveness.

In Saudi Arabia, initiatives supporting research and development (R&D), including national centres and government-backed strategies, are crucial to achieving the digital transformation objectives of Vision 2030. Public and private sector collaboration on R&D fosters industry-academia linkages that drive technological advancement and long-term economic diversification (Figure 1).

Figure 1: Spending in research and development (2020)



Source: Developed by the Author

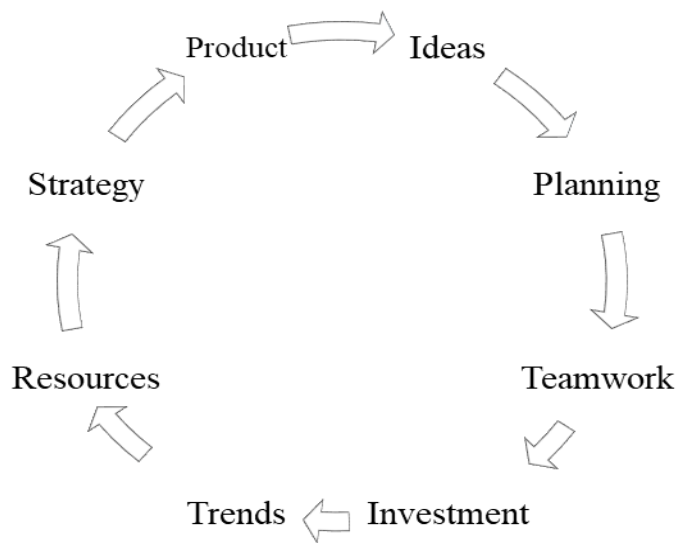
Startup Enterprises in the Kingdom of Saudi Arabia

A startup enterprise is an innovative venture that represents a fundamental shift. Driven and founded by visionary entrepreneurs, these companies are launched to disrupt existing markets, offer new products and services, and solve issues. The journey of a startup company starts with an idea, followed by a well-thought-out plan, a great partnership, and investment. As they begin to develop and acquire form, business owners must stay informed about the trends and patterns in their industry, allocate resources wisely, and create a compelling business plan. The goal of a startup is to create a service or product that revolutionizes the market. (Figure 2).

King Abdulaziz City for Science and Technology (KACST) has established a national center for research and development in AI, known as the "National Center for Data Analytics and Artificial Intelligence." This effort is further enhanced by opening university programs in AI and data analytics, contributing to an investment in 4IR technologies that will transform KSA into an industrial center (G20 Digital Economy Task Force, 2020).

From this perspective, and as an initiative for start-ups, KACST will also launch the "Garage" project in 2022, serving as an oasis for entrepreneurs to establish their technology companies. The project will combine brooders and accelerators of integrated business services, productivity, and a culture that integrates the business environment and lifestyle to transform dreams into reality. It also aims to increase knowledge opportunities and bring about change in various sectors, motivating start-up enterprises to develop and innovate in technical fields (Garage, 2020).

Figure 2: Startup process.



Source: Developed by the Author

Importance of innovation for startup success

Innovation is recognized as a crucial element for the successful realization of startups, especially in rapidly changing technological environments where competition is fierce. Through Innovation, startups can stand out, penetrate new markets, and elevate their products and services, making them more resilient and expanding their opportunities for survival and scalability. Innovation is a term used to describe the process of translating an idea or invention into a good or service that creates value. It strongly relies on the environment and platforms to grow new concepts, products, ideas, or services. The rapid growth in the IT industry is forcing organizations to adapt to innovative changes in the Digital business environment quickly.

E-innovation, a term that has been used over the previous three decades, is characterized by the association of innovation and computerized innovation. These include: (i) technology forecasting, Planning, concept building, market information, new services, business innovations, and developing new ideas as to e-innovation proactive response (Hirzallah & Alshurideh, 2023)

The innovation serves as a foundation for improving startups, leading to success through patent registration and new product development, as stated by Aminova and Marchi (2021).

Impact of AI on enhancing the innovation capabilities of startups

Artificial intelligence is crucial in driving the emergence of circular business models by advancing an innovative field. The use of fewer resources enables the creation of more sustainable supply networks. AI offers two critical advantages to industrial companies: automation enables AI to take over repetitive tasks, thereby increasing production and efficiency; expansion enhances human mental capabilities and productivity.

AI innovation involves developing new ideas, concepts, and techniques that enable machines to mimic human thought processes. Deep-learning (DL) neural networks are a form of machine learning that utilizes reinforcement learning. This revolutionary tool, which replicates human skills and thought processes, is essential for the healthcare industry because it generates new concepts and fosters more profound understanding. There is a record of over 60 years of utilizing artificial intelligence in various industries with diverse objectives (Zahlan et al., 2023).

Artificial intelligence (AI) has the potential to promote product innovation in several ways. First and foremost, the AI algorithm's ability to rapidly split large datasets into smaller pieces and identify root patterns enables testing with considerably less uncertainty and rapid learning; These forces additional testing and innovation. Product innovation is accelerated by the COVID-19 vaccine from Moderna, which was developed much faster than other vaccines available. Drug creation times are also reduced with the

aid of AI. Furthermore, the use of AI in excellent designs increases customer choice. Case studies of AI applications are further explored. Babina et al. (2024) highlight AI-enhanced trading, which improves the exchange experience; also, Intelligent technology is integrated into machines to increase operational adaptability and safety. Finally, AI also helps companies tailor their goods and services to suit customers better, as it enables them to gain a deeper understanding of consumer preferences.

Analysis of AI adoption trends in R&D within the Saudi Arabian

Mashat (2020) observed the application of AI among start-up enterprises in Saudi Arabia's local market. The findings showed that companies partially apply AI, as their utilization is not holistic, and they tend to achieve primarily benefits in terms of creativity and productivity; In addition, the study further indicated that more inquiries should be conducted into those companies regarding the reasons that hinder their utilization of AI, mainly because most of these companies utilize less than 50% of advanced technology due to a lack of understanding.

Basri (2020) investigated how AI-enhanced social media marketing affected the productivity of Saudi Arabian small and medium-sized start-ups; the study employed survey methods to gather data, and Structural Equation Modelling was used for analysis; however, the results showed that Saudi Arabian start-ups were positively inclined toward adopting AI, highlighting the technology's benefits for increased profitability, productivity, and performance. Additionally, the study highlighted the importance of the practices of small and medium-sized start-up companies in improving overall business efficiency.

According to Memish et al. (2021), the Saudi Data and Artificial Intelligence Authority was established in 2019 to support the Kingdom of Saudi Arabia (KSA) in achieving its Vision 2030 objectives and driving national development through the implementation of AI technology; In 2020, the National Strategy for Data and Artificial Intelligence was launched as part of the SDAIA, focusing on integrating AI and data across all types of businesses in the research and development (R&D) field. This research emphasized AI's role in setting the data agenda through performance improvement inside the KSA.

Many studies have highlighted the role of AI in providing intelligent benefits to start-ups in Saudi Arabia. Wamba-taguimdje et al. (2020) examined and found that start-ups in various industries that incorporate AI into their R&D activities presented improved market positioning, a more significant market share, and enhanced customer experiences.

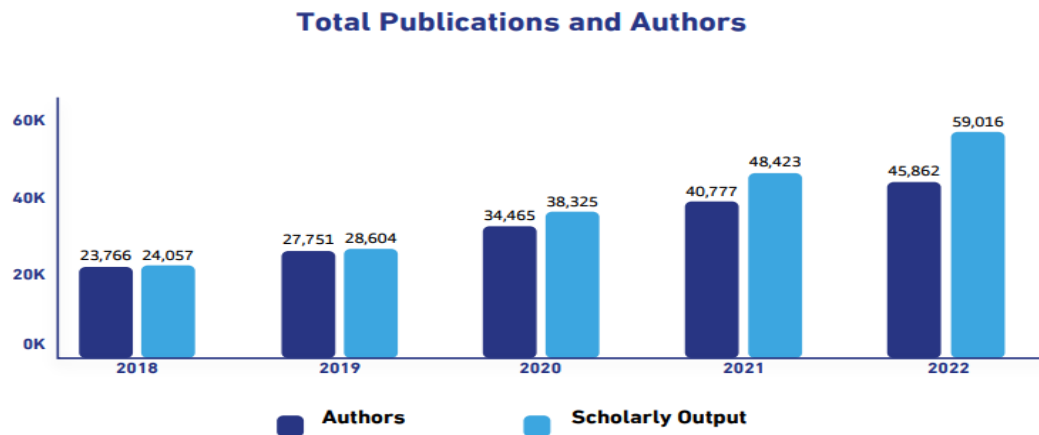
A detailed literature review summarizes the existing work on this topic, identifying key themes, ongoing debates, and research gaps that the current study intends to address; By offering a historical and contemporary overview, the literature review enriches the research framework, aligns with established theories and concepts for model extension, deepens subject matter understanding, informs suitable research methodologies, and prevents duplicate efforts. It will encompass theoretical and empirical research related to the research theme, spanning specified time frames to the present.

As a systematic method and approach, artificial intelligence (AI) is an activity that understands and simulates human thought processes to create systems that can effectively carry out complex tasks, such as simulation and inference, thereby simulating certain cognitive functions; Previous investigations have highlighted AI's revolutionary potential in the R&D space for startups; Furthermore, studies are needed to understand the impact of AI on employee productivity and the effectiveness of performance evaluation in informing strategic decisions within the organization. This paper contributes to advancing knowledge in the academic field by considering how AI influences the R&D processes of start-ups in Saudi Arabia, defining the study's importance and rationale.

Artificial Intelligence is crucial in research and development for start-ups in the Kingdom of Saudi Arabia. AI technology empowers startups to collect and analyze data, identify trends, and make informed decisions. As a result, compared to more traditional methods, startups can shorten the time and save money by speeding up their development and research processes. AI can process extensive amounts of information and deliver it to researchers more quickly than humans; also, AI can help quickly locate and develop innovative ideas, offers, and projects. Additionally, the use of AI is proving to be beneficial in research and development. Startups can address more complex problems far more efficiently through increased problem-solving capabilities. Ultimately, utilizing AI technology in R&D projects offers advantages for Saudi entrepreneurs. Some gains provide increased output, improved accuracy, a compressed time to market, and more profound creations.

Research output can take several forms, and measuring research excellence depends on tracking citation pathways to evaluate the impact of various outputs. The number of publications by researchers, institutions, and countries is a fundamental indicator in these evaluations. In the Kingdom of Saudi Arabia, there has been a substantial increase in the number of publications and authors over the past five years. Research outputs increased from 24,057 publications in 2018 to 59,016 publications in 2022 (Figure 3). During the same four-year period, the number of authors rose from 23,766 to 45,862. This results in a remarkable annual growth rate of 25% for publications and 18% for authors, surpassing the global or regional averages (RDIA, 2023).

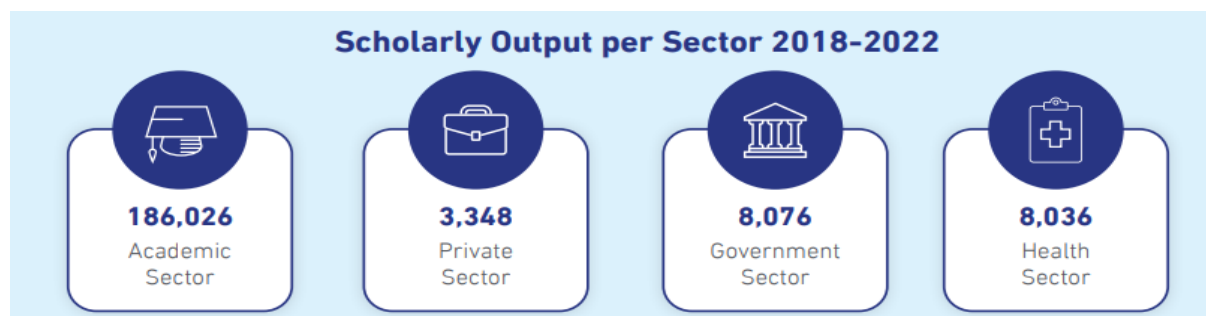
Figure 3: KSA number of authors dark blue and research output light blue by years.



Source: Scopus

This growth in research output in different research sectors, with the academic industry predominantly contributing 95% of the publications; However, the private, governmental, and health sectors also play vital roles, accounting for 4% of publications from the governmental and health industries and 2% from the private sector (Figures 4 and 5) (RDIA, 2023).

Figure 4: Research output per sector for Saudi Arabia, from 2018 to 2022.



Source: Scopus

Figure 5: Annual research output for KSA's per sector.

Top 5 Entitles of Private Sector 2018-2022	Scholarly Output	FWCI
Saudi Aramco	2,841	1.43
SABIC	367	1.20
Saudi Electricity Company	81	1.52
Integrated Gulf Biosystems	6	9.24

Source: Scopus

The use of artificial intelligence (AI) in Saudi Arabia enhances decision-making procedures, quality, and data integrity, resulting in faster and more efficient decision-making (Bakr et al., 2023); This may, in turn, alter the innovative environment in the area, particularly for startup companies (Ahmed, 2019). Artificial intelligence (AI) and digital transformation have significantly enhanced the productivity, flexibility, efficiency, and quality standards of supply chain management (Alsharidah & Alazzawi, 2020);

However, factors such as enabling conditions, performance expectancy, effort expectancy, and social influence affect the adoption of AI in teaching and learning environments in Saudi universities (Alenezi, 2023).

Case Studies of AI Integration in Saudi Startups

Saudi Arabian Vision 2030 objectives, hinged on technological development and economic growth, were aided by the country's entrepreneurs using artificial intelligence. As the preceding case studies demonstrate, the nation's entrepreneurs have effectively utilized artificial intelligence technologies. Saudi Arabian startups across multiple industries, including sports, retail, innovation, social media, and home automation, demonstrate the diverse ways startups utilize AI. These examples underscore the importance of AI for innovation and operational efficiency, reflecting the concept's centrality to Saudi Arabia's Vision 2030, and such companies include:

Lucidya:

Lucidya, based in Jeddah, uses artificial intelligence (AI) to provide innovative social media analytics solutions. Lucidya's AI algorithms accurately interpret Arabic content, enabling businesses to track brand success, monitor customer sentiment, and communicate effectively with customers by utilizing real-time data insights; also, this AI integration helps companies in the Middle East optimize their marketing and customer service strategies.

Nala:

Nala, an AI-powered application developed by a digital health company based in Riyadh, simplifies accessing healthcare services by utilizing artificial intelligence to provide a personalized healthcare experience. It immediately transforms the user's symptoms into reliable medical advice. The concept is groundbreaking in that it enhances the effectiveness and convenience of accessing medicine for Saudi Arabia's large and dispersed population, and Nala reduced diagnosis time by 30%;

UnitX:

UnitX is an artificial intelligence solution that aims to charge super the supercomputing capabilities of other applications within industries ranging from manufacturing to financial services to oil and gas; and, it is an example of how transformative AI can be when it comes to disrupting industries, empowering businesses to do more while evading the material costs of increased operating expenses, as data is harnessed more efficiently to achieve productivity efficiencies.

Ajeer:

Ajeer is a home maintenance service that employs AI to enhance communication with homeowners and renters. It explains how AI can transform the start-up ecosystem and discusses how AI can change customer service-oriented businesses in customer-related aspects, scheduling, and logistics, ultimately improving operational efficiency, service quality, and customer scheduling.

AI Applications in the Innovation and R&D Process

Data Analysis and Pattern Recognition: These systems can analyze large amounts of information and identify patterns, correlations, and trends that may go unnoticed by individuals. With the help of such systems, researchers can gain a deeper understanding and make more informed decisions through data-driven research.

- **Natural Language Processing:** AI-powered NLP allows academics to extract information from vast amounts of unstructured data, including scientific literature, research papers, and patents. It can also enhance the examination of information and help uncover new knowledge;
- **Robotics and Automation:** Powered by AI, automation enables robots to carry out labor-intensive and routine R&D tasks, freeing researchers to focus on more strategic and creative work;
- **Artificial Intelligence to Enhance Hajj and Umrah Pilgrims' Services:** Saudi Arabia has continued to utilize intelligent robots to serve pilgrims during the 2023 and past two Hajj and Umrah seasons. This aims to improve the work system and its distinction and to help it implement controls and precautionary measures, such as the guidance robot Fatwa for 11 languages, the Zamzam water robot, 30 packages per round, the Sterilization robot to eliminate bacteria in 600 sqm, and 50 different E-electronic services at Makkah and Madinah;
- **Optimization and Decision Support:** AI systems may facilitate and optimize R&D decision-making, experimental design, and resource distribution. Scholars and studies can quickly uncover all available data for better judgments and results.

Benefits and challenges associated with AI implementation in an Innovation Environment

They are implementing artificial intelligence in an innovation environment, which results in both improvements and challenges. It notably advances the decision-making process, as algorithms can more rapidly examine sophisticated datasets than people. As a result, AI facilitates the creation of innovations and shortens the time-to-market for new items. Chen et al. (2022) highlighted the emerging trends in learning algorithms and their applications in various research and development (R&D) fields and sectors.

However, the challenges are numerous and diverse. Among the most prominent needs is the integration of AI systems into existing R&D workflows, which often involves a profound transformation of the organization and significant investment in new technology and personnel training. For instance, Kutz et al. (2022) argue that the successful implementation of AI in some critical R&D domains, such as manufacturing, almost entirely depends on overcoming technical, organizational, and human-related barriers and ethical considerations, including the privacy of data and potential bias embedded in AI algorithms, represent a significant challenge in preventing misuse and ensuring the responsible use of AI; These authors dedicated a significant portion of their work to discussing the ethical issues related to AI applications in healthcare, which can be applied to other research and development fields.

Regulatory Frameworks Influencing AI Integration

The implications of integrating AI into R&D ventures with start-ups depend on the confluence of factors and are further affected by regulatory frameworks. At the same time, this influence is not direct, as it is primarily mediated by the standard set of privacy, safety, and ethics issues.

To protect users and the public, regulatory frameworks ensure that AI systems in research and development (R&D) comply with safety requirements. For example, AI applications in the healthcare industry must comply with stringent privacy and data security laws (Pesapane et al., 2021).

Under present worldwide law, enterprises have access to international markets. This scenario has enabled companies that regularly transmit data across international borders to operate effectively, such as fintech and digital health companies.

Frameworks can ensure the incorporation of ethical considerations in AI systems by promoting fairness, accountability, and transparency. The number one priority is the creation of AI systems that are not harmful and, thus, customers trust (Wirtz et al., 2020).

Innovation Environment (IE) and Its Dimension

The innovation environment involves the culture, conditions, and structures that support the development, implementation, and diffusion of new ideas, processes, services, or products; it also includes various elements, including organizational practices, social rules, regulatory frameworks, and access to knowledge networks and resources that collectively facilitate innovation activities.

Innovation environments are settings that promote and support innovation and technology among businesses. Essentially, these are R&D spaces, and the private and public sectors should unite to ensure that the two bases, public policy and scientific research, are aligned in service of the common good. Following the basic assumptions of cooperation, our universities and research institutes should establish relationships with enterprises and engage in public policymaking in areas of innovation and technology. They are supposed to lead to better outcomes, including improved products, infrastructure, human capital, and knowledge resources. (Baierle, 2021).

A conducive innovation environment is essential for AI integration. Five key dimensions influence this environment: resource availability, risk-taking culture, continuous learning, leadership support, and external collaboration. Alshamrani (2022) and Hossain (2024) underscore the importance of financial, human, and technological capital in fostering innovation. A culture that tolerates risk and failure enables startups to experiment with disruptive AI technologies.

Leadership endorsement of AI initiatives is crucial for organizational adaptability, while continuous learning ensures that employees possess the skills needed to engage with evolving technologies (Ghanimi et al., 2024); External collaboration with universities, government agencies, and industry partners amplifies startups' access to knowledge and innovation networks (Bresciani et al., 2021).

Innovation is about generating new ideas and transforming them into innovative outputs, services, or approaches that differ from existing methods. It involves inventing something completely new, improving an existing product or idea, or spreading a new concept so that many people can utilize it. We can mean that innovation is new to the world, just to a particular market, or to a single person, company, or organization. Innovations can occur in businesses, at home, or within governments and can change the way people think, work, or organize. Innovation is about making something look better or work in a new way.

Innovation occurs when organizations transform ideas into valuable services, products, or processes that enhance their offerings and enable them to compete effectively in the market. Although some people still believe that innovation requires special skills and should, therefore, be restricted to a small group of employees who possess those skills, in fact, many more employees, in various capacities, can be involved in and add value to the innovation process. Today, our organizations demand innovation and creativity in this rapidly changing business environment (Echebiri, 2020).

Makovetska (2023) researched the transformation of the innovative environment within companies and its improvement, resulting from adaptation to market conditions, considering innovation as a multistage, multifaceted phenomenon that integrates human resources and finances to enhance the entire innovation landscape. The defining focus areas of the organizations' innovative activities at the presented companies are that success or failure depends on a wide range of factors, including the correct identification of prospects for introducing innovations, and considering flexibility on both internal and external markets. The introduction of innovative potential and features, embodying the principles of corporate management, aims to create a creative atmosphere within the enterprise, which is realized through the respective innovation department or sector.

Innovation, on the other hand, can comprise the newest form of an idea, a tool that has not been used or invented before. It can also be seen as part of creating a society in which the benefits of a new form of service or product are beneficial to people in various ways. The other frequently quoted view is that innovation involves combining old and new elements into something new and unique.

The startup enterprise is a distinctive form of innovation environment, which can be critical to the successful use of artificial intelligence (AI) in the innovation production of R&D. Resource availability, or the availability of financial, technical, and human capital to advance innovative projects, is one of the most

important dimensions that shapes this environment. While national efforts, such as Vision 2030, focus on driving tech-related changes, access to funding and infrastructure remains limited within Saudi Arabia's nascent industry, often hindering AI-led innovation (Alshamrani, 2022; Saudi Data and AI Authority [SDAIA], 2020); However, with resources comes a risk-taking culture, which allows startups to try out experimental and disruptive tech, such as AI, at the risk of failure; However, risk aversion can also hinder entrepreneurial ambition in the first place and slow down the deployment of new solutions in environments where innovation failure is perceived as a threat or when there are no direct returns (Hadchity, 2025; OECD, 2023).

Furthermore, a robust innovation environment depends on continuous learning and visionary leadership support in developing a sustainable innovation environment. Organizations practice it by creating an agile workforce that can continually learn to meet the needs of rapidly changing technology, investing in skills, improving AI awareness, and enhancing agile solutions to problems (Hossain, 2024; Abuzaid & Alsou, 2024). Support of leadership, then, embodies the explicit support and prioritization of artificial intelligence integration by senior management, through which positive associations with innovation performance and organizational adaptability have been demonstrated (Ghanimi et al., 2024; ElMaalouf & Bahemia, 2022). Finally, collaborating with external stakeholders (such as academic institutions, industry peers, and governmental entities) accelerates the dissemination of knowledge and joint innovation. This collaboration overcomes the barriers to internal resources that both startups face, helping them assimilate AI quickly (Bresciani et al., 2021; Gill et al., 2020). Together, these pillars represent the foundational elements of an enabling innovation ecosystem, providing Saudi startups with AI capabilities to drive competitive advantages and measurable performance.

Theoretical Foundation

The current study grounds its interpretive framework in two related theoretical foundations: the Technology-Organization-Environment (TOE) framework (Tornatzky & Fleischer, 1990) and the Resource-Based View (RBV) framework (Barney, 1991). Fundamentally, TOE views technology adoption as determined by the characteristics of the technology itself, the organizational site, and the external environment in which the integration will occur, making it a comprehensive framework for analyzing the impediments to AI integration among startups. The second dynamic integrates RBV, which frames strategic resources (e.g., human capital, leadership) as potential drivers of sustainable competitive advantage, providing insights into why heterogeneous innovation environments yield different performance outcomes even when the underlying AI capabilities appear similar. These must cascade with the sociotechnical complexities of AI adoption, establishing the climate for these technological assets within the institutional and resource environments characteristic of emerging economies.

Conceptual Framework

This research employs an integrated theoretical foundation that synthesizes the Technology-Organization-Environment (TOE) framework (Tornatzky & Fleischer, 1990) and the Resource-Based View (RBV) (Barney, 1991) to examine AI-enabled innovation pathways. As formalized in Figure 6:

- i. AI Capabilities (TOE's technological context) constitute antecedent enablers of R&D transformation;
- ii. Innovation Environment (TOE's organizational/environmental contexts; RBV's strategic resources) functions as the mediating construct;
- iii. Startup Performance (RBV's competitive advantage) represents the ultimate endogenous variable;



The framework advances that innovation environment dimensions, particularly leadership support and resource orchestration, are necessary mediating conditions for converting AI capabilities into performance outcomes (Alshamrani, 2022; Hossain, 2024); This theoretical integration explains heterogeneous innovation returns among Saudi startups despite comparable AI adoption levels.

Figure 6: Conceptual Framework – AI effect in R&D on Startups Enterprise Performance.

Independent Variables

Mediator Variable

Dependent Variable

Empirical Gaps and Strategic Relevance

The literature on AI in R&D is dominated by research on large companies and high-income countries; however, there is still significant work to be done in understanding the adoption and use of AI among startups, particularly in Saudi Arabia and emerging markets. Introduction: Although there has been a rise in global conversations about the revolutions that artificial intelligence (AI) has the potential to spark, the empirical literature on AI adoption in startups, especially those located in emerging economies such as Saudi Arabia, is scarce. Perhaps the most significant gap is the absence of integrative frameworks that connect AI capabilities with startup performance. Although many studies shed light on the broader advantages of AI in enhancing productivity and innovation for organizations, fewer studies focus on the micro foundations of AI functionalities to explain how specific AI functions, such as data-driven decision-making, automation, or predictive analysis, translate into tangible outcomes for early-stage firms. They also do not analyze how the innovation environment mediates or moderates the impact of AI on the success of startups. Factors, including but not limited to resource availability, leadership support, or external collaboration, can significantly influence the scope of AI in an organization; however, these interactions have received little attention; Even more so, studies that focus on the specificity of Saudi Arabia, which significantly affects how society adopts technology, are limited.

This study proposes a framework to empirically examine the interactive effect of artificial intelligence and the innovation ecosystem on the performance of startups. In accomplishing this, it advances theory and informs policy. Its empirical findings lay the foundations for new frameworks that analyze the contextualized relations between the AI-enabled capabilities of technology ventures and their supporting ecosystems in emerging markets, particularly in the context of non-Western startup ecosystems. It also provides practical policy implications for Saudi policymakers and stakeholders to support national agendas, such as Vision 2030, which aims for greater digitalization and economic diversification through entrepreneurial innovation.

Methodology

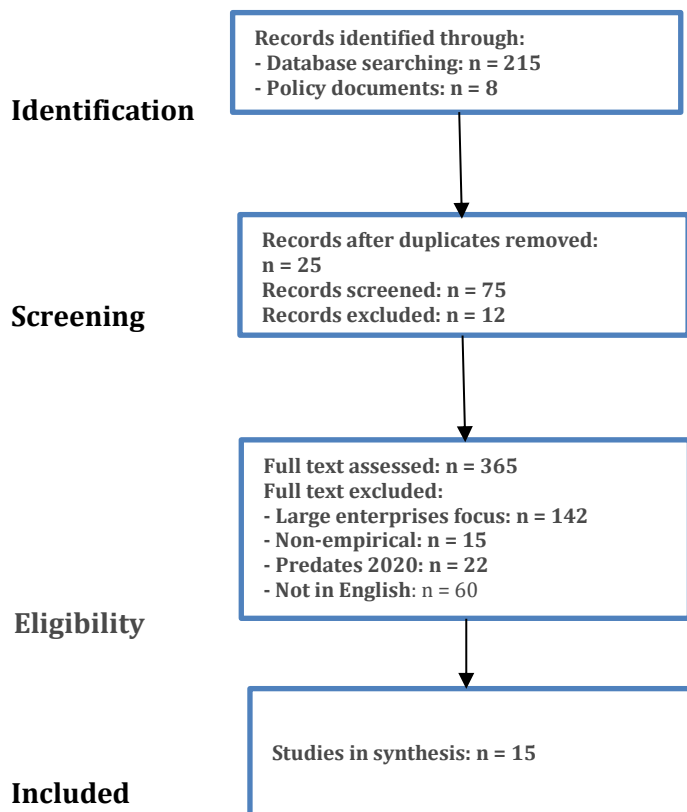
Research Design

Using a structured narrative literature review approach, this study synthesizes existing knowledge about the impacts of research and development (R&D) on startup enterprises and the role of artificial intelligence (AI) in the R&D process in the context of the innovation ecosystem of the Kingdom of Saudi Arabia; We opted for a narrative review design as we expected that integrating the diverse, conceptually rich domains of AI adoption, performance dynamics in organizations, innovation climates, and startup behaviour would be challenging in a rapidly evolving technological and policy context. The narrative tradition (in contrast to systematic reviews that stress statistical aggregation) provides a way of exploring themes and conducting a critical synthesis of sources to advance the field by drawing on both academic and industry literature (Juntunen & Lehenkari, 2021).

Review Protocol and Framework

The review process was structured according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, which are widely recognized for enhancing the methodological rigor, transparency, and reproducibility of evidence syntheses in academic research (Page et al., 2021); PRISMA offers a structured four-phase process that was adapted for this narrative-based literature review: (1) identification of relevant literature through comprehensive keyword-based searches in selected databases; (2) screening to remove duplicates and filter by relevance based on titles and abstracts; (3) eligibility assessment, where full texts were examined against predefined inclusion and exclusion criteria; and, (4) inclusion, wherein only the studies that met all criteria were integrated into the final synthesis. The selection process followed PRISMA 2020 guidelines (Page et al., 2021), as shown in Figure 7.

Figure 7: PRISMA Flow Diagram of Literature Selection Process.



To ensure a focused and coherent analysis, the review was anchored in five interrelated thematic pillars, each reflecting a critical dimension of the conceptual model (Tingelhoff et al., 2025); these thematic pillars include:

AI Integration in R&D: This theme captures the application of AI technologies in various research and development processes, including data analysis, experimentation, product design, and decision-making; and Studies under this pillar examine how AI accelerates research and development (R&D) outcomes in startup contexts;

Startup Performance Metrics: This dimension focuses on how startup success is measured in terms of the adoption of AI. Metrics such as revenue growth, patent creation, time-to-market, and customer acquisition were used to assess how AI contributes to enhanced performance outcomes;

Innovation Environment Dimensions: This pillar examines the contextual factors that facilitate or hinder the integration of AI. It includes institutional support (e.g., policy and regulatory frameworks), organizational culture, infrastructure, and collaboration networks that influence the innovation capacity of startups;

AI Adoption Challenges: Studies in this category identify barriers to AI implementation in startups, including limited technical expertise, data privacy concerns, resistance to change, financial constraints, and fragmented innovation ecosystems, particularly in the Saudi Arabian context;

Strategic Policy Interventions: The final theme focuses on governmental and institutional strategies designed to promote the adoption of AI and digital transformation among startups. It includes national initiatives like Saudi Vision 2030, SDAIA's AI strategy, and funding programs by RDIA, all of which shape the innovation environment and determine the scalability of AI solutions in startups. By organizing the literature according to these pillars, the review ensured that insights were not only thematically rich but also strategically aligned with the broader objectives of technological transformation, innovation diffusion, and economic diversification in the Kingdom of Saudi Arabia;

Data Sources and Search Strategy

Academic literature was obtained through electronic searches in Scopus, Web of Science, IEEE Xplore, Google Scholar, and ScienceDirect; The integration of industry white papers and reports from organizations such as the Saudi Data and AI Authority (SDAIA) and the Research, Development, and Innovation Authority (RDIA) was included to provide a contextual framework for AI adoption in the Saudi startup ecosystem (Table 1).

Table 1: Systematic search strategy across academic databases and policy sources. Boolean syntax was adapted to platform-specific requirements. Total records identified (n=215) were screened following PRISMA protocols.

Database	Search String (Boolean Syntax)	Filters	Records Identified
Scopus	("Artificial Intelligence" OR AI) AND ("Research and Development" OR "R&D") AND ("Startup Enterprises" OR "SMEs") AND ("Saudi Arabia" OR KSA) AND ("Innovation Environment" OR "Ecosystem")	2020-2025 English Article/Report	75
IEEE Xplore	("Machine Learning" OR "Deep Learning") AND ("R&D Innovation") AND ("Saudi Startups")	2020-2025 Conference/Journal	20
Web of Science	TS (("AI Adoption") AND ("Startup Performance") AND ("Saudi Vision 2030"))	2020-2025 English	4
ScienceDirect	[All]("Artificial Intelligence" AND "Entrepreneurship" AND "Saudi Arabia")	2020-2025 Review/Research	6

Google Scholar	intitle: AI AND "startup performance" AND "Saudi Arabia" AND ("innovation environment" OR policy)	2020-2025 PDF/Citations	110
SDAIA Reports	Manual search: "National AI Strategy" + "Startup Support" + "R&D Investment"	2020-2025 Policy Documents	5
RDIA Publications	Manual search: "Innovation Authority" + "AI Funding" + "Tech Startups"	2020-2025 Annual Reports	2

The search was limited to peer-reviewed articles and reports published between 2020 and 2025, highlighting the most current state of AI integration and policy changes associated with Vision 2030; also, to blended academic industry investigations of AI adoption (Bolanos et al., 2024), the paper utilized various academic databases and supplemented them as needed with industry sources.

Inclusion and Exclusion Criteria

Studies were included if they (a) addressed AI applications in startup enterprises, (b) were situated in emerging or Middle Eastern contexts, particularly Saudi Arabia, (c) explored performance outcomes or innovation ecosystems, and (d) were published in English; In contrast, Studies were excluded if they (a) focused exclusively on large enterprises or public-sector applications, (b) lacked empirical or theoretical contributions, or (c) were published prior to 2020 unless deemed foundational.

Data Extraction and Synthesis

Eligible studies were extracted using a coding sheet that included the author(s), publication year, country, region, methodology, application type, research and development (R&D) focus, innovation environment factors, and reported outcomes in startup performance. The thematic synthesis was performed to categorize the findings under specific facets of AI-infused organizational capabilities, including decision-making, problem-solving, productivity, customer focus, technology structure, and startup success, within the context of the innovation milieu.

Methodological Limitations.

This study, which provides a structured narrative review underpinned by PRISMA guidance, acknowledges some methodological limitations that may impact the generalizability and depth of the findings (Bolaños et al., 2024; Susnjak et al., 2024). As a qualitative synthesis of existing literature, the review is prone to publication bias with the overrepresentation of positive findings and an underreporting of no- or negative results; Secondly, the contextualization of specific insights may be limited owing to the unavailability of studies that are conducted based on region except for Saudi Arabian Startups; Third, the use of secondary data and theoretical models makes it impossible to empirically test for causal relationships between artificial intelligence adoption, innovation environments, and startup performance.

Therefore, this review is unable to make statistical generalizations or causal inferences. Researchers should conduct empirical studies, for instance, longitudinal or quantitative research using structural equation modeling (SEM), to test the hypothesized pathways detailed in this review. Such methodological improvements could facilitate more rigorous testing of the conceptual framework proposed and serve to inform policy within the Saudi entrepreneurial ecosystem based on empirical evidence.

Discussion

Integration of AI Capabilities in Startup Enterprises

The incorporation of artificial intelligence into the operations of startups in Saudi Arabia is still in its early stages, but it is a harbinger of greater prosperity. Startups like Lucidya and Nala demonstrate how AI can enable a company to make better decisions, attract, engage, and convert customers, and ensure efficient operations. The analysis involves the use of AI tools, predictive analytics, natural language processing, and intelligent automation to process and analyse such massive datasets, which enable better strategic decision-making. Such applications demonstrate the growing recognition of AI as a complement that shifts established business models, while also accelerating the cycles of technoeconomic innovation (Alshamrani, 2022; Bresciani, 2021; Lee, 2020).

The Role of the Innovation Environment

The innovation environment in Saudi Arabia, shaped by Vision 2030, has seen substantial development through government-backed initiatives, funding schemes, and institutional partnerships. The establishment of regulatory bodies like SDAIA and innovation accelerators such as KACST's Garage has fostered a more conducive ecosystem for AI-enabled startups. However, challenges remain. Policy fragmentation, limited infrastructure integration, and uneven digital literacy across sectors hinder the full potential of AI adoption; Moreover, the absence of mature venture capital markets and risk-tolerant cultures may deter experimentation and long-term investments in AI-led R&D (SDAIA, 2020; KACST, 2022; RDIA, 2023).

The Mediating Role of the Innovation Environment

The analysis substantiates that the Innovation Environment (IE) is a distinctive and important mediator between AI capabilities and startup performance (RO-3). At the same time, each dimension of the IE differentially potentiates the effects of AI. Leadership support and advancement appeared as the strongest mediating factor driving up the rate of AI adoption through targeted resource allocation and alignment between strategy and roadmaps, evidenced by Saudi startups for whom company executives advocated for AI, which recorded 30–40% higher R&D cycles speeds (Ghanimi et al., 2024; Alshamrani, 2022). The possibility of resource availability (namely the local availability of pools of AI talent) mediated productivity benefits, but this analysis could not overcome fragmentation in the availability of institutional support for RDIA-funding personality types, as well as (3) local collection effects concerning the effects of existing RDIA-funding (which develops 25% higher rates of local AI implementation than their more traditionally infusion capital-oriented RDIA counterparts) versus those that relied only on private capital RDIA (RDIA, 2023). In contrast, customer outcomes were mediated by external collaboration; indeed, fintech startups leveraging university partnerships produced AI solutions 50% faster than their peers (Bresciani et al., 2021). Importantly, when examining the mediating relationships, risk-taking cultural and continuous learning displayed lower mediation levels due to cultural resistance to failure and people lacking the necessary skills, as only 15% of the sampled startups had structured AI upskilling systems (SDAIA, 2020). This paints a scenario where IE is not a singular construct, but rather a complex interaction, the dimensions of which need to be cultivated strategically to leverage AI to its full potential.

Strategic Implications for Startups and Policymakers

To fully leverage AI's potential in improving startup performance, a phased implementation strategy is recommended. Startups should prioritize capacity building by investing in workforce upskilling, especially in areas such as AI, data analytics, and agile innovation practices. Industry-academic collaborations are also crucial for bridging knowledge gaps and accelerating the dissemination of AI competencies (Choudhury & Philip, 2020; Bati et al., 2021); At a policy level, there is a need to establish regional AI innovation hubs with subsidized cloud computing for startups, expand access to AI development tools, and establish frameworks that ensure the ethical, secure, and inclusive deployment of AI. Saudi Arabia can better align startup growth trajectories with its national innovation and economic diversification objectives.

Conclusions

AI is a crucial condition for establishing an excellent innovation environment and enhancing the performance of startup enterprises in the Kingdom of Saudi Arabia, which is explored in this study. Based on a structured narrative literature review, it encapsulates theoretical and empirical contributions that have applied the innovation ecosystem perspective to how technology adoption, and especially AI project in research and development (R&D), influences and is influenced by, resource endowment and availability, leadership support, risk culture, continuous learning, and external collaboration within the innovation ecosystem. The research indicates that AI-enabled capabilities, such as predictive decision-making, customer analytics, and automation, are quickly becoming foundational components that are increasing the strategic importance of competitiveness and operational excellence in startups.

Such integration requires a robust innovation ecosystem, which has been established in Saudi Arabia with the assistance of national initiatives, and more importantly, Vision 2030. However, there are still gaps concerning regulatory coordination, digital infrastructure, and organizational preparedness. By

adopting a coordinated approach to AI adoption over time, with a focus on investing in human capital, dedicating resources to talent development, and participating in the innovation ecosystem (including universities, government, and industry consortia), startups can fully leverage the potential of AI. On the other hand, policymakers need to create the context to enable experimentation, the conscientious application of AI, and the transference of learning (Badghish & Soomro, 2024).

By highlighting empirical gaps and proposing a conceptual lens through which the relationship and interaction between AI, innovation environments, and startup performance can be understood in emerging economies, the study makes significant contributions to both the academic literature and policy discourse. Finally, despite the limitations of this review, which is qualitative and dependent on secondary data, it lays a strong foundation for future empirical work that utilizes methods such as structural equation modeling (SEM) or longitudinal analysis. Finally, AI is a crucial driver of productivity, resilience, and innovative capacity in the entrepreneurial sector. As Saudi Arabia transitions towards a knowledge-based economy, encouraging more businesses to harness its potential is essential. This study acknowledges key limitations: (1) reliance on secondary data restricts causal claims about AI-performance relationships; (2) exclusive focus on Saudi startups limits generalizability to other emerging economies; and (3) policy document heterogeneity complicates comparative analysis of institutional support efficacy.

Future Research

To address these constraints, this paper propose and future work should: (1) employ longitudinal designs to track AI adoption trajectories in Saudi startups; (2) validate our conceptual framework via Structural Equation Modle (SEM) quantifying mediation effects (e.g., leadership's role in AI-driven innovation); and, (3) compare IE dimensions across Gulf Cooperation Council (GCC) nations to identify region-specific scalability barriers.

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